The Turning Effect — Mark schemes

PAPERS A TO D · 30 MARKS EACH · TUTOR COPY

Marks in square brackets show how the total is split. Award method marks for a correct formula and a correct substitution even when the arithmetic fails. Two marks are lost more often than any others on this topic: the missing unit conversion from centimetres to metres, and the missing direction on an answer. The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

1 (a) The turning effect of a force about a pivot. [3]
(b) moment = force × perpendicular distance from the pivot.
(c) N m (newton metre).

The word "perpendicular" is required in (b). Without it, do not award the mark.

2 (a) 20 × 0.5 = 10 N m [4]
(b) 25 cm = 0.25 m [1]; 40 × 0.25 = 10 N m [1]
(c) 12 × 1.5 = 18 N m

In (b), 40 × 25 = 1000 scores zero. The conversion is the whole point of the question.

3 A longer handle gives a larger distance d from the pivot [1] [2]
Since moment = F × d, the same force produces a larger moment, so the nut turns more easily [1]

4 (a) Clockwise [2]
(b) Anticlockwise

5 (a) 0.6 m [2]
(b) 2.5 m

6 40 cm = 0.40 m [1] [3]
25 × 0.40 = 10 N m [1]
Clockwise [1]

Omitting the direction costs a mark for a word the student already knew.

7 The handle is far from the hinges, which are the pivot, so d is large [1] [2]
A larger distance gives a larger moment for the same push, so the door opens more easily [1]

8 300 × 2 [1] [2]
= 600 N m [1]

9 When an object is balanced, the total clockwise moment about a pivot [1] [2]
equals the total anticlockwise moment about the same pivot [1]

"About the same pivot" is worth insisting on — it becomes essential on Papers C and D.

10 (a) The hinges [2]
(b) The nut (the centre of the nut)

11 (a) 15 × 0.8 = 12 N m anticlockwise [4]
(b) 24 × 0.5 = 12 N m clockwise
(c) The two moments are equal [1], so the beam balances [1]

12 The distance must be measured at right angles to the line of action of the force [1] [2]
If the force is at an angle, only the perpendicular part of the distance produces turning, so using the full distance overestimates the moment [1]

Paper B — Medium

30 MARKS

1 Total clockwise moment = total anticlockwise moment [1] [2]
about the same pivot, when the object is balanced [1]

2 Ravi's moment = 320 × 1.6 = 512 N m [1] [3]
400 × d = 512 [1]
d = 1.28 m [1]

Heavier child, shorter distance. If the answer comes out bigger than 1.6 m, it is wrong before it is checked.

3 (a) Distance = 50 − 20 = 30 cm = 0.30 m [1]; 2 × 0.30 = 0.6 N m [1] [4]
(b) 4 × d = 0.6, so d = 0.15 m = 15 cm [1]; on the other side of the pivot, so at the 65 cm mark [1]

Using 20 cm as the distance instead of 30 cm is the standard error. Distances are measured from the pivot, not from the end of the rule.

4 Effort moment = 200 × 1.2 = 240 N m [1] [3]
W × 0.2 = 240 [1]
W = 1200 N [1]

5 (a) 250 × 1.2 = 300, and 150 × 2.0 = 300 [1]; total = 600 N m [1] [4]
(b) 400 × d = 600 [1], so d = 1.5 m [1]

6 (a) Near the hinges the distance from the pivot is small [1], so the same force gives a much smaller moment and the door resists opening [1] [4]
(b) The load's distance from the wheel (the pivot) is smaller [1], so the load's moment is smaller and less force is needed at the handles to balance it [1]

7 That child's distance from the pivot decreases [1] [3]
so their moment decreases and the moments are no longer equal [1]
The see-saw tips down on the other side [1]

8 45 cm = 0.45 m [1] [2]
18 × 0.45 = 8.1 N m [1]

9 Any two sensible examples [2] — spanner, door handle, crowbar, wheelbarrow, scissors, bicycle pedal [3]
One explained using a large distance from the pivot producing a large moment for a small force [1]

10 N is a unit of force, not of moment [1] [2]
A moment must be given in N m, because it is a force multiplied by a distance [1]

Paper C — Hard

30 MARKS

1 (a) The weight of a uniform plank acts at its centre, 2 m from the left end [1]; that is 2 − 1 = 1 m from the pivot [1] [4]
(b) 200 × 1 = 200 N m [1], acting clockwise (the centre lies to the right of the pivot) [1]

2 (a) The beam is uniform, so its weight acts at its centre, which is exactly where the pivot is [1]. The distance from the pivot is zero, so the moment is zero [1] [4]
(b) 20 × 1.2 = 24 N m; 30 × d = 24 [1], so d = 0.8 m [1]

3 5 × 0.4 = 2 N m [1] [4]
3 × 0.9 = 2.7 N m [1]
Total anticlockwise = 4.7 N m [1]
F × 0.5 = 4.7, so F = 9.4 N [1]

Moments on the same side add. Students who subtract them have confused "same side" with "opposite side".

4 At an angle, the perpendicular distance from the pivot to the line of action is shorter than the handle itself [1] [2]
Since the moment uses that shorter perpendicular distance, the moment is smaller [1]

5 150 N at each support [1] [3]
The beam is uniform, so its weight acts at the centre, which is the same distance from each support [1]
By symmetry the two supports share the 300 N equally [1]

6 The plank is not uniform [1] [3]
Its weight acts at the balance point, 0.8 m from the left end [1]
A uniform plank would have balanced at its centre, 1.0 m from the end, so more of its mass lies towards the left [1]

7 (a) Distance = 40 − 10 = 30 cm = 0.30 m; 2 × 0.30 = 0.6 N m [1] [5]
(b) The rule’s weight acts at 50 cm, which is 0.10 m to the right of the pivot [1]; 1 × 0.10 = 0.1 N m clockwise [1]
(c) Clockwise needed = 0.6 − 0.1 = 0.5 N m [1]; the 90 cm mark is 0.5 m from the pivot, so W × 0.5 = 0.5 and W = 1 N [1]

Forgetting the rule’s own weight in (c) gives 1.2 N. That omission is what this question exists to catch.

8 The lighter child sits further from the pivot [1] [2]
so that weight × distance is the same on both sides, even though the weights differ [1]

9 (a) 30 ÷ 0.25 = 120 N [3]
(b) 30 ÷ 0.40 = 75 N
(c) A longer spanner needs a smaller force for the same moment [1]

The moment required is fixed by the nut, not by the person. Only the trade between force and distance changes.

Paper D — Hard

30 MARKS

1 (a) 250 × 1.6 = 400 and 200 × 0.8 = 160 [1]; total = 560 N m [1] [5]
(b) 300 × 1.2 = 360 N m [1]
(c) The fourth child must supply 560 − 360 = 200 N m [1]; W × 2.0 = 200, so W = 100 N [1]

2 (a) 600 × 1.5 = 900 N m anticlockwise [1] [5]
(b) The weight acts at the centre, 2 m from the left end, which is 0.5 m to the right of the pivot [1]; 400 × 0.5 = 200 N m clockwise [1]
(c) Clockwise still needed = 900 − 200 = 700 N m [1]; the right end is 2.5 m from the pivot, so F = 700 ÷ 2.5 = 280 N [1]

Two common failures: forgetting the beam’s weight entirely (giving 360 N), and using 4 m instead of 2.5 m as the distance to the right-hand end.

3 (a) 250 × L = 400 [1], so L = 1.6 m [1] [4]
(b) Any sensible problem, explained [2] — it becomes unwieldy or too heavy to handle; she may not be able to reach the far end while keeping the pivot in place; the bar may bend or snap.

4 It must be the perpendicular distance [1] [3]
measured from the pivot to the line of action of the force [1]
It matters whenever the force is not at right angles to the arm — then the perpendicular distance is shorter than the distance to your hand, and the student's version overestimates the moment [1]

When the force is perpendicular, the student's statement happens to give the right answer, which is why the misunderstanding survives so long.

5 The bus pivots about the outer wheels on the bend [1] [4]
Passengers upstairs have their weight acting higher up and further out as the bus leans [1]
This gives a larger perpendicular distance from the pivot, so a larger overturning moment [1]
When the overturning moment exceeds the restoring moment of the bus's own weight, it tips [1]

6 (a) Load moment = $500 \times 0.4 = 200$ N m [1]; $F \times 1.2 = 200$ [1]; $F = 167$ N (166.7 N) [1] [5]
(b) Moving the load closer reduces its distance from the wheel [1], so its moment falls and a smaller force at the handles balances it [1]

Accept 166.7 N or 167 N. The wheel is the pivot — students who use the far end of the barrow lose the first mark.

7 True, but only if the distance stays the same [1] [4]
Moment = $F \times d$, so doubling F doubles the moment when d is unchanged [1]
If the distance changes at the same time, it does not: for example, doubling the force while halving the distance leaves the moment unchanged, since $2F \times d/2 = Fd$ [2]

A worked numerical example is fully acceptable in place of the algebra.